

ZeroPaper: An Autonomous Research System

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ABSTRACT

ZeroPaper is an autonomous system that takes a research domain as input and produces a publication-candidate manuscript as output, with no human intervention between launch and completion. The system runs ten numbered stages and six adversarial gates, coordinates roughly thirty specialized agents, and produces finished LaTeX for economics at roughly two dollars per paper amortized under a flat-fee Claude Code Max subscription (approximately zero marginal cost per additional paper).

From ten premises about how LLMs behave and what they cost, we derive six design principles for long-running autonomous systems: keep state and control flow outside the model, spend context carefully, delegate aggressively, verify rather than trust, make termination mechanical, and run independent work in parallel. ZeroPaper runs on Claude Code, OpenAI Codex, and Gemini CLI, with variants for finance and macro and extensions for empirical analysis and LLM experiments.

DETAILS

SERIES	IAR Methodology Papers
NUMBER	IAR-M-001
POSTED	11 May 2026
DRAFTED	1 May 2026
KEYWORDS	autonomous research; AI for science; LLM agents; adversarial evaluation; system design; Markov systems
STABLE URL	instituteforautomatedresearch.org/papers/iar-m/iar-m-001
DOI	10.5281/zenodo.20127843
ISSN	Pending (IAR Methodology Papers)

HOW TO CITE

Lopez-Lira, Alejandro, ZeroPaper: An Autonomous Research System (May 1, 2026). IAR Methodology Papers No. IAR-M-001, Institute for Automated Research, Available at:
<https://instituteforautomatedresearch.org/papers/iar-m/iar-m-001>,
<https://doi.org/10.5281/zenodo.20127843>

BIBTEX

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@techreport{lopezlira2026zeropaper,  
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  title       = {{ZeroPaper}: An Autonomous Research System},  
  institution = {Institute for Automated Research},  
  type        = {IAR Methodology Papers},  
  number      = {IAR-M-001},  
  year        = {2026},  
  month       = {May},  
  url         = {https://instituteforautomatedresearch.org/papers/iar-  
m/iar-m-001},  
  doi         = {10.5281/zenodo.20127843}  
}
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